📝 Unit 8 Guided Notes — Transformations

ANSWER KEY

Slide it. Flip it. Turn it. Resize it. Then describe what changed and what stayed the same.

🎯 By the end of this unit, I can…

- Perform translations, reflections, rotations, and dilations on polygons
- Describe a sequence of transformations between two figures
- Tell congruence from similarity based on the transformations used

Words to know

- **Transformation** — a move or change of a shape.
- **Pre-image** — the shape before the move.
- **Image** — the shape after the move. Its points get a prime mark: A′.
- **Rigid transformation** — a move that keeps the same size and shape.
- **Congruent** — same size and same shape.
- **Similar** — same shape. The size can be different.
- **Scale factor (k)** — the number you multiply by in a dilation.

Standards: NC.8.G.2 · NC.8.G.3 · NC.8.G.4

Answer key: every blank and final answer is filled in green.

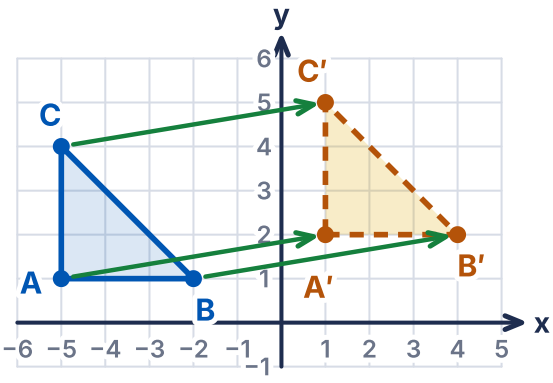
1 Transformation Vocabulary

Before we slide, flip, turn, or resize anything, we need shared language. Every transformation moves points by a rule — and we always describe what changed and what stayed the same.

📌 **Transformation:** A rule that moves, flips, turns, or resizes every point of a figure in a predictable way.

📌 **Pre-image:** The *original* figure, BEFORE the transformation. Vertices use plain letters: A, B, C.

📌 **Image:** The *new* figure, AFTER the transformation. Vertices use a prime mark: A′, B′, C′ (read "A prime").

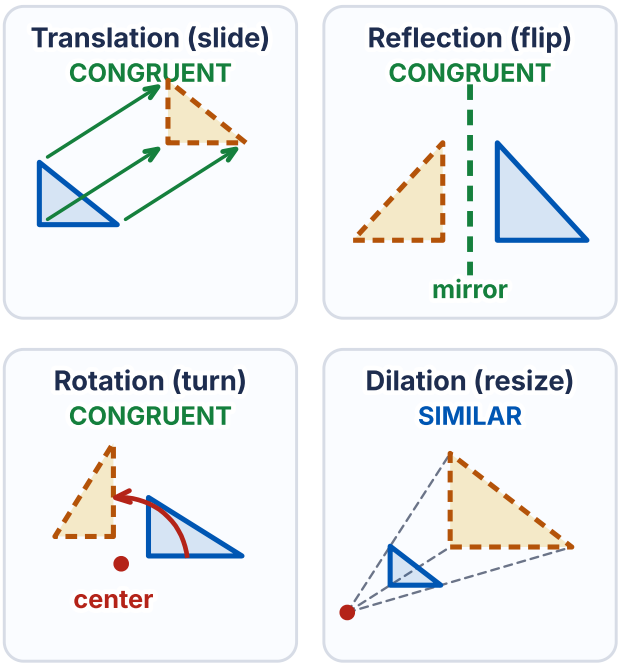


Pre-image ABC → **image A′B′C′**. Each point A goes to its partner A′.

Two Families of Transformations

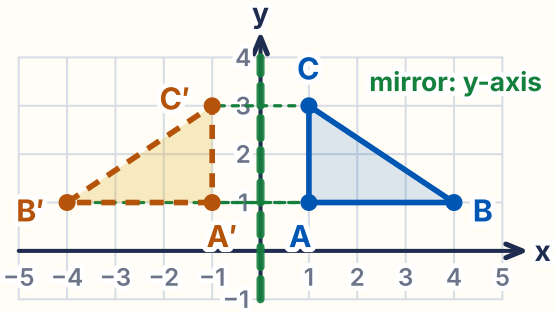
Family	What Stays the Same	Types	Result
Rigid (size stays the same)	Size AND shape	Translation, Reflection, Rotation	Pre-image & image are **CONGRUENT**
Non-Rigid	Shape only (size changes)	Dilation	Pre-image & image are **SIMILAR**

💡 **Big idea:** If size stays the same, it's **congruent**. If size changes but shape stays the same, it's **similar**.



Quick Check

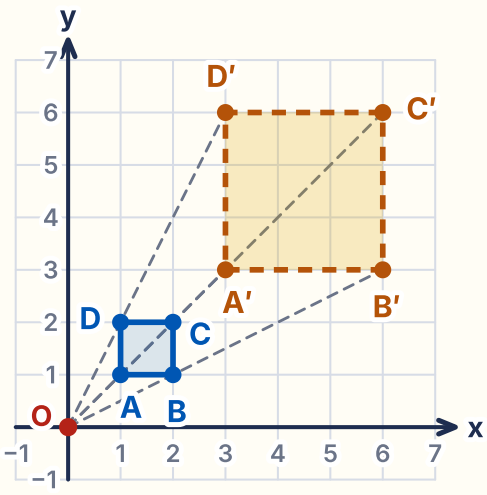
Q1 A triangle is reflected over the y-axis. The pre-image and image are...



blue = pre-image · **gold dashed** = image

- ○ Similar but not congruent
- ● **Congruent**
- ○ Neither

Q2 A square is dilated by a scale factor of 3. The pre-image and image are...



Rule: (x, y) → (3x, 3y)

blue = pre-image · **gold dashed** = image

- ● **Similar but not congruent**
- ○ Congruent
- ○ Neither

2 Translations (Slides)

📌 **Translation:** Every point slides the *same distance* in the *same direction*. The shape is NOT rotated or flipped — it just moves.

The Rule

$$(x,\ y) \rightarrow (x + h,\ y + k)$$

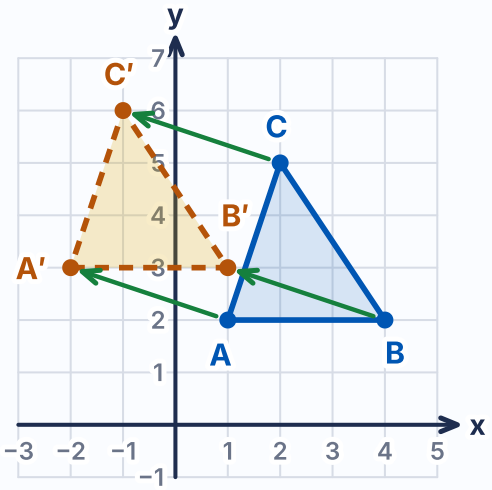
h = horizontal shift | **+** moves right, **−** moves left
k = vertical shift | **+** moves up, **−** moves down

💡 **Tip:** Add the numbers. + moves right or up. − moves left or down.

Worked Example

Example 1: Triangle ABC has vertices A(1, 2), B(4, 2), C(2, 5). Translate by (x − 3, y + 1).

Pre-image △ABC (blue).



Rule: (x, y) → (x − 3, y + 1)

blue = pre-image · **gold dashed** = image

Step 1 — Identify h and k: h = −3 (shift left 3), k = +1 (shift up 1)

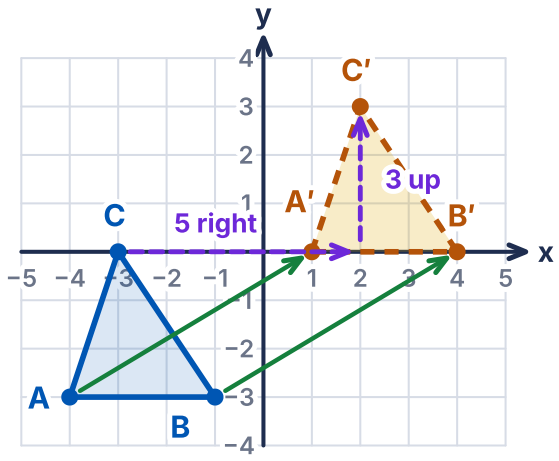
Step 2 — Apply rule to A(1, 2): (1 + (−3), 2 + 1) = (−2, 3) → A′(−2, 3)

Step 3 — Apply rule to B(4, 2): (4 + (−3), 2 + 1) = (1, 3) → B′(1, 3)

Step 4 — Apply rule to C(2, 5): (2 + (−3), 5 + 1) = (−1, 6) → C′(−1, 6)

✓ **Image vertices: A′(−2, 3), B′(1, 3), C′(−1, 6)**

Visual



Rule: (x, y) → (x + 5, y + 3)

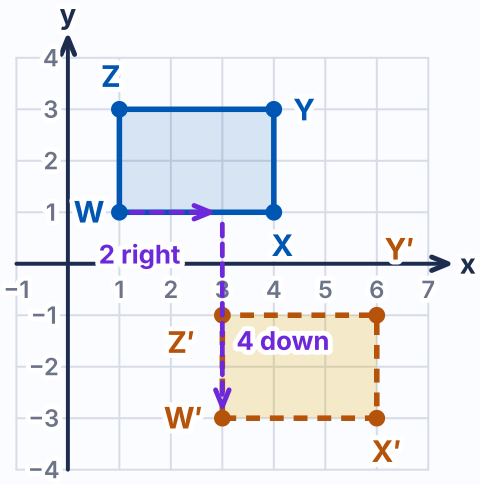
blue = pre-image · **gold dashed** = image

Rule (x + 5, y + 3): **h = +5** moves every point 5 right, **k = +3** moves it 3 up. Every arrow is the same slide.

Practice

P1 **Rectangle WXYZ has vertices W(1, 1), X(4, 1), Y(4, 3), Z(1, 3). Translate the rectangle by the rule (x + 2, y − 4). What are the coordinates of W′?**

- ● **(3, −3)**
- ○ (−1, 5)
- ○ (3, 5)
- ○ (2, −4)

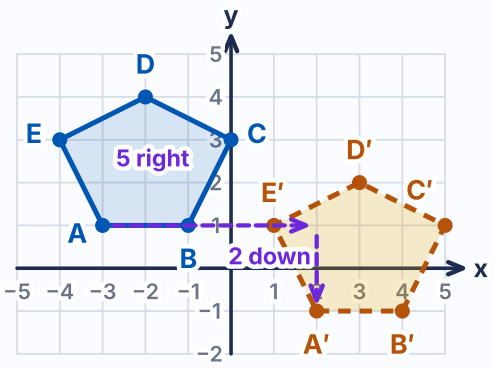


Rule: (x, y) → (x + 2, y − 4)

blue = pre-image · **gold dashed** = image

P2 **Pentagon ABCDE has vertex A(−3, 1). The whole pentagon is translated by (x + 5, y − 2). Write the coordinates of A′.**

A′ = (2 , −1)

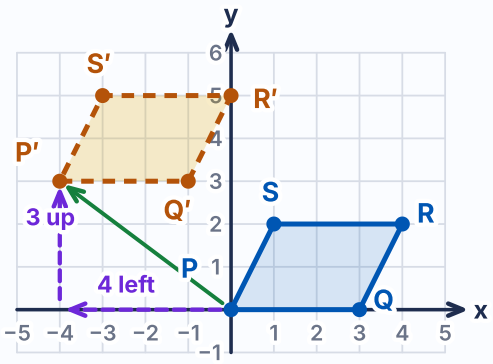


Rule: (x, y) → (x + 5, y − 2)

blue = pre-image · **gold dashed** = image

P3 **Parallelogram PQRS was translated to P′Q′R′S′ (shown). P(0, 0) maps to P′(−4, 3). Which rule was used?**

- ● **(x − 4, y + 3)**
- ○ (x + 4, y − 3)
- ○ (x − 3, y + 4)
- ○ (x + 4, y + 3)



Rule: (x, y) → (x − 4, y + 3)

blue = pre-image · **gold dashed** = image

3 Reflections (Flips)

📌 **Reflection:** Every point flips across a **line of reflection** (the mirror line). Each point lands the same distance from the mirror, on the opposite side.

📌 **NC Standard 8.G.3:** 8th grade focuses on reflections across the **x-axis** and **y-axis**.

💡 **One more mirror line:** To reflect over the line **y = x**, swap the numbers: **(x, y) → (y, x)**. Example: (1, 4) → (4, 1).

The Rules

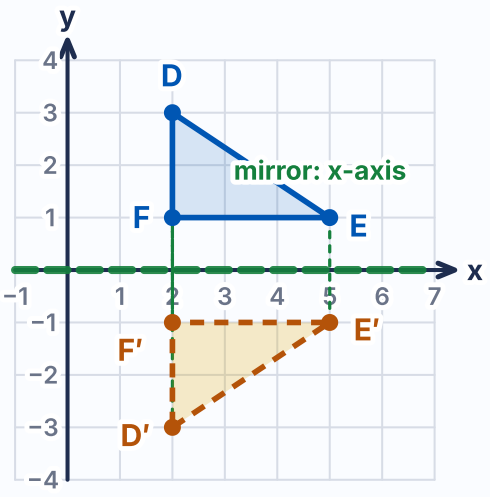
Line of Reflection	Rule	Example: (3, 4) →	What Changes
x-axis	`(x, y) → (x, −y)`	(3, −4)	y flips sign
y-axis	`(x, y) → (−x, y)`	(−3, 4)	x flips sign

📌 **Memory trick:** The coordinate that *matches the axis name* is the one that **stays the same**. Reflect over the x-axis → x stays. Reflect over the y-axis → y stays.

Worked Example

Example 2: Reflect triangle DEF with D(2, 3), E(5, 1), F(2, 1) over the x-axis.

Pre-image △DEF (mirror = x-axis).



Rule: (x, y) → (x, −y)

blue = pre-image · **gold dashed** = image

Step 1 — Identify the rule: Reflection over x-axis: (x, y) → (x, −y). Keep x; flip the sign of y.

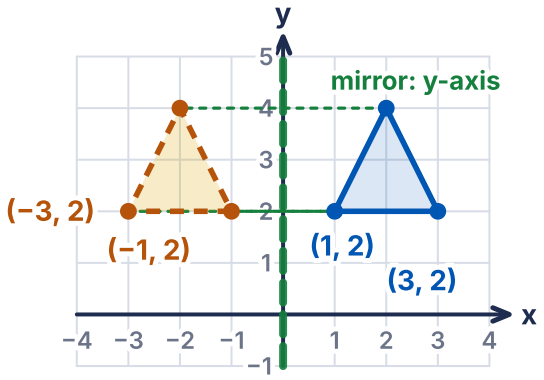
Step 2 — Apply to D(2, 3): (2, −3) → D′(2, −3)

Step 3 — Apply to E(5, 1): (5, −1) → E′(5, −1)

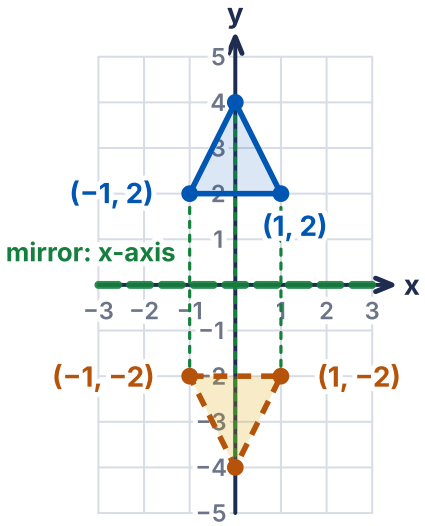
Step 4 — Apply to F(2, 1): (2, −1) → F′(2, −1)

✓ **Image vertices: D′(2, −3), E′(5, −1), F′(2, −1)**

Visual — Two Mirrors Side by Side



Reflect over the y-axis: x flips sign



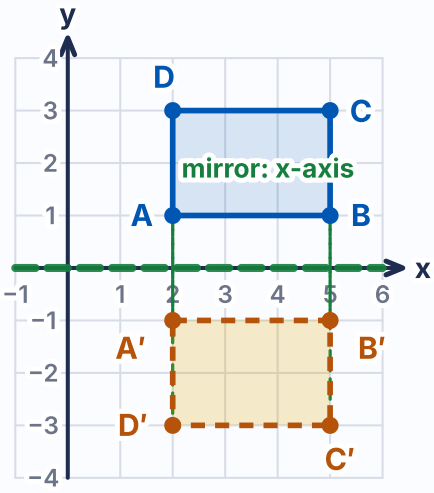
Reflect over the x-axis: y flips sign

blue = pre-image · **gold dashed** = image

Practice

P1 Rectangle ABCD has vertex C(5, 3). The whole rectangle is reflected over the x-axis. What are the coordinates of C′?

- ● **(5, −3)**
- ○ (−5, 3)
- ○ (−5, −3)
- ○ (3, 5)

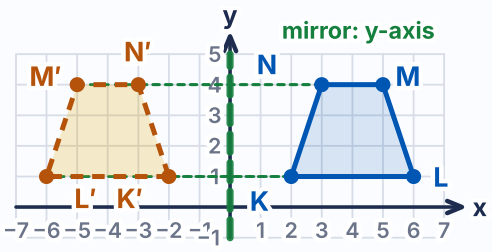


Rule: (x, y) → (x, −y)

blue = pre-image · **gold dashed** = image

P2 Trapezoid KLMN has vertex M(5, 4). The whole trapezoid is reflected over the y-axis. Write the coordinates of M′.

M′ = (**−5** , **4**)

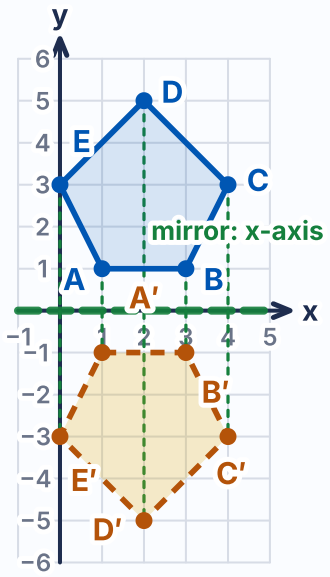


Rule: (x, y) → (−x, y)

blue = pre-image · **gold dashed** = image

P3 Pentagon ABCDE was reflected to A′B′C′D′E′ (shown). Which line is the mirror (line of reflection)?

- ● **the x-axis**
- ○ the y-axis
- ○ the line y = x
- ○ the line x = 2



blue = pre-image · **gold dashed** = image

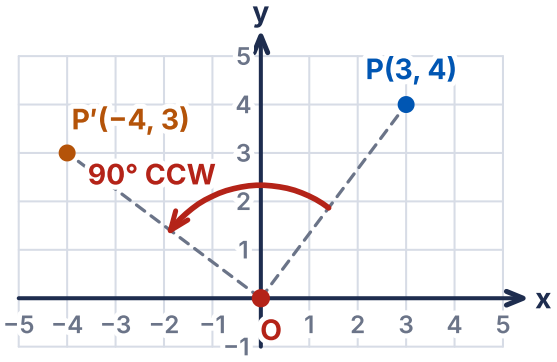
4 Rotations (Turns)

📌 **Rotation:** Every point turns a fixed angle around a **center of rotation**. In this unit, the center is always the **origin (0, 0)**.

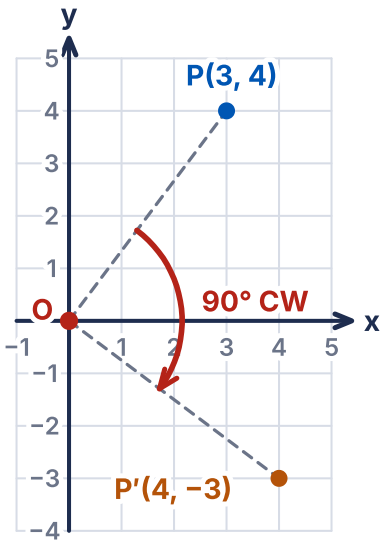
📌 **NC Standard 8.G.3:** 8th grade focuses on rotations of **90°, 180°, and 270°** about the origin.

↻ **Direction matters:** A rotation can be **counterclockwise (CCW)** or **clockwise (CW)**. Always check which way the problem asks. **270° one way = 90° the other way.** 180° is the same either way.

The Rules (About the Origin)



90° counterclockwise



90° clockwise

Same point, same 90° turn about O, opposite directions. The red arrow shows the direction.

Learn these three rules.

Rotation	Rule	Example: (3, 4)
90° CCW	$(x, y) \to (-y, x)$	(−4, 3)
180°	$(x, y) \to (-x, -y)$	(−3, −4)
90° CW	$(x, y) \to (y, -x)$	(4, −3)

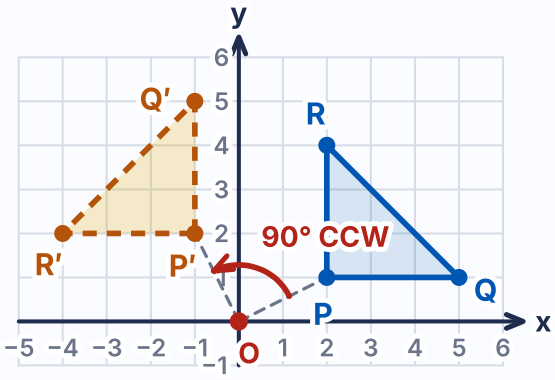
📌 **270°?** 270° CCW is the same as 90° CW. 270° CW is the same as 90° CCW.

⚠ **Watch out:** 180° is the only rotation where the coordinates are NOT swapped. For 90° and 270°, you must swap x and y first, THEN apply the sign rule.

Worked Example

Example 3: Rotate triangle PQR with P(2, 1), Q(5, 1), R(2, 4) 90° counterclockwise about the origin.

Pre-image △PQR and the 90° counterclockwise turn about the origin.



Rule: (x, y) → (−y, x)

blue = pre-image · **gold dashed** = image

Step 1 — Identify the rule: 90° CCW about origin: (x, y) → (−y, x)

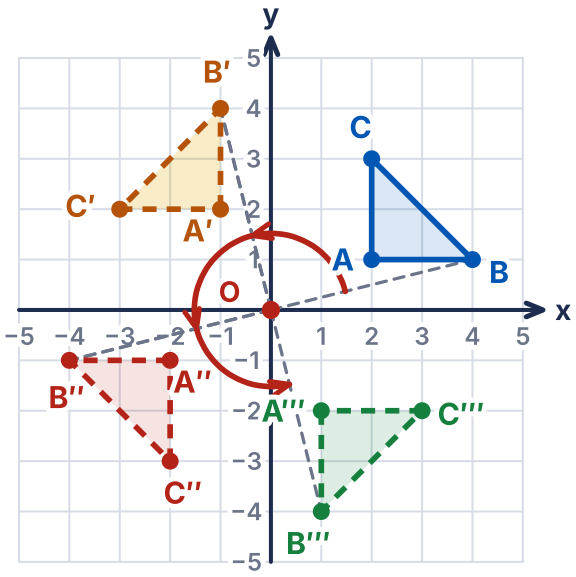
Step 2 — Apply to P(2, 1): (−1, 2) → P′(−1, 2)

Step 3 — Apply to Q(5, 1): (−1, 5) → Q′(−1, 5)

Step 4 — Apply to R(2, 4): (−4, 2) → R′(−4, 2)

✓ **Image vertices: P′(−1, 2), Q′(−1, 5), R′(−4, 2)**

Visual — One Triangle, Three Rotations



↻ each red arrow = one 90° CCW turn

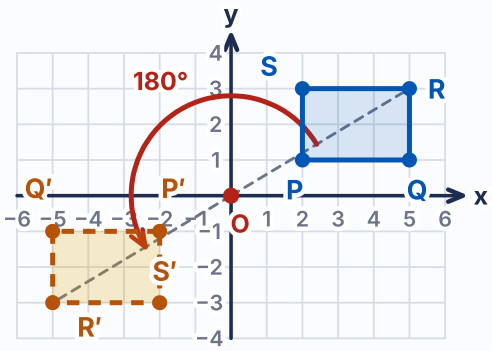
△ABC → 90° CCW → 180° → 270° CCW (= 90° CW). Each red arrow is one more 90° turn about O.

Same triangle △ABC, three different rotations. Notice the triangle stays the same size and shape — only where it sits and which way it faces change.

Practice

P1 Rectangle PQRS has vertex R(5, 3). The whole rectangle is rotated 180° about the origin. What are the coordinates of R′?

- ● **(−5, −3)**
- ○ (5, −3)
- ○ (−3, −5)
- ○ (3, 5)

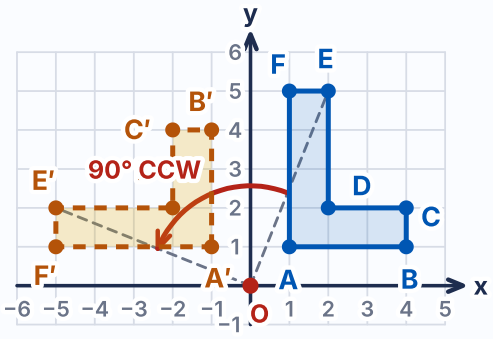


Rule: (x, y) → (−x, −y)

blue = pre-image · **gold dashed** = image

P2 **L-shape ABCDEF has vertex E(2, 5). The whole shape is rotated 90° counterclockwise (CCW) about the origin. Write the coordinates of E′.**

E′ = (**−5** , **2**)

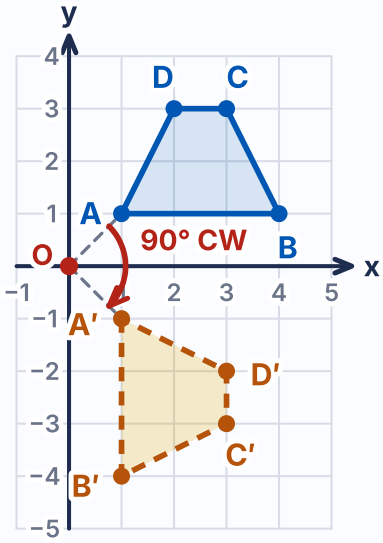


Rule: (x, y) → (−y, x)

blue = pre-image · **gold dashed** = image

P3 **Trapezoid ABCD was rotated to A′B′C′D′ (shown). All rotations are about the origin. Which rotation was used?**

- ● **90° clockwise (CW)**
- ○ 90° counterclockwise (CCW)
- ○ 180°
- ○ 270° clockwise (CW)



blue = pre-image · **gold dashed** = image

5 Dilations (Resize)

📌 **Dilation:** A transformation that **resizes** a figure by a **scale factor k** from a center point (usually the origin). The shape stays the same; the size changes.

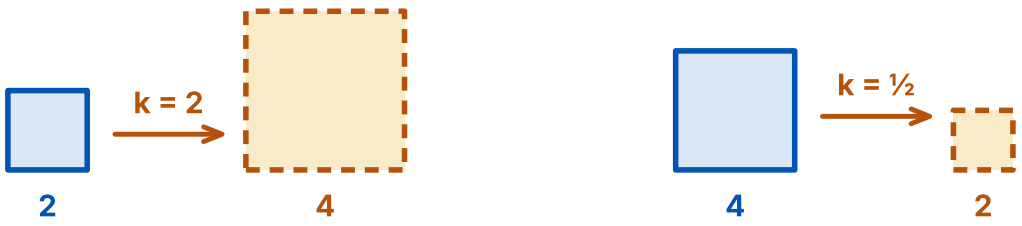
📌 **NC Standard 8.G.4:** Dilations in 8th grade are centered at the origin. Scale factor k can be a whole number, decimal, or fraction.

The Rule (Centered at the Origin)

$$(x, y) \rightarrow (kx, ky)$$

Multiply *both* coordinates by the scale factor **k**.

What k Tells You



k > 1 → bigger (enlargement) · 0 < k < 1 → smaller (reduction) · k = 1 → same size

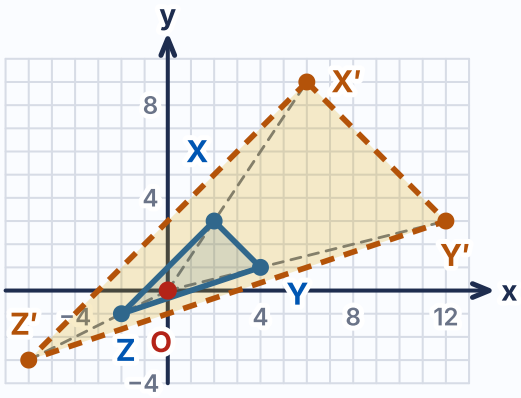
Scale Factor k	Effect	Image vs. Pre-image
k > 1	Enlargement	Image is **bigger**
k = 1	No change	Same size
0 < k < 1	Reduction	Image is **smaller**

⚠ **Watch out:** Dilation is the ONLY transformation in this unit that is NOT rigid. The pre-image and image are SIMILAR, but NOT congruent (unless k = 1).

Worked Example

Example 4: Triangle XYZ has X(2, 3), Y(4, 1), Z(−2, −1). Dilate by scale factor k = 3 (centered at origin).

Pre-image △XYZ with rays from the center (0, 0).



Rule: (x, y) → (3x, 3y)

blue = pre-image · **gold dashed** = image

Step 1 — Identify the rule: (x, y) → (3x, 3y). Multiply both coordinates by 3.

Step 2 — Apply to X(2, 3): (3 · 2, 3 · 3) = (6, 9) → X′(6, 9)

Step 3 — Apply to Y(4, 1): (3 · 4, 3 · 1) = (12, 3) → Y′(12, 3)

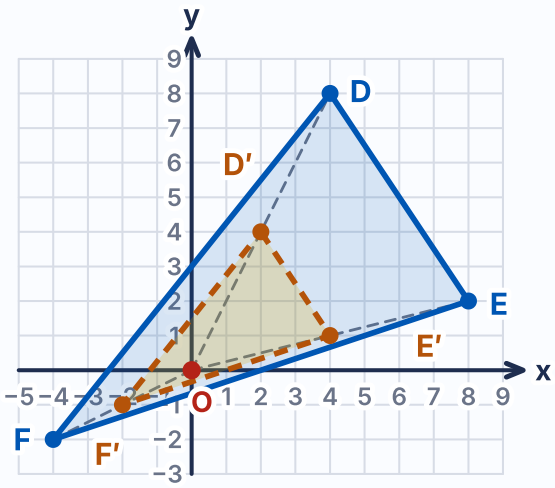
Step 4 — Apply to Z(−2, −1): (3 · (−2), 3 · (−1)) = (−6, −3) → Z′(−6, −3)

✓ **Image vertices: X′(6, 9), Y′(12, 3), Z′(−6, −3) — bigger by a factor of 3**

Worked Example — Reduction

Example 5: Triangle DEF has D(4, 8), E(8, 2), F(−4, −2). Dilate by scale factor k = $\frac{1}{2}$ (centered at origin).

Pre-image △DEF with rays from the center (0, 0).



Rule: (x, y) → (½x, ½y)

blue = pre-image · **gold dashed** = image

Step 1 — Rule: (x, y) → ($\frac{1}{2}$)x, ($\frac{1}{2}$)y). Multiply both coordinates by $\frac{1}{2}$ (or divide by 2).

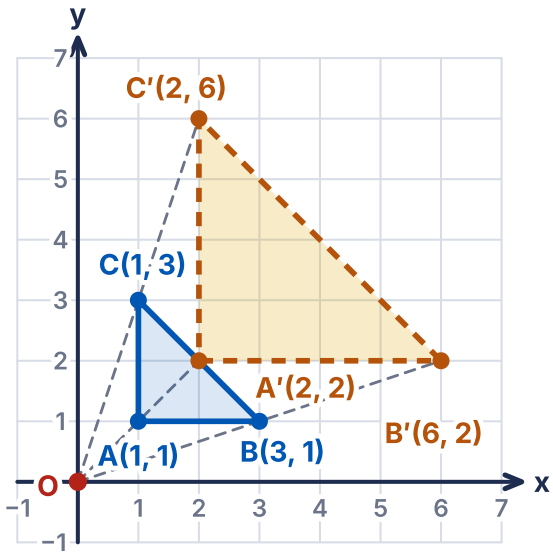
Step 2 — Apply to D(4, 8): ($\frac{1}{2}$)(4), ($\frac{1}{2}$)(8)) = (2, 4) → D′(2, 4)

Step 3 — Apply to E(8, 2): ($\frac{1}{2}$)(8), ($\frac{1}{2}$)(2)) = (4, 1) → E′(4, 1)

Step 4 — Apply to F(−4, −2): ($\frac{1}{2}$)(−4), ($\frac{1}{2}$)(−2)) = (−2, −1) → F′(−2, −1)

✓ Image vertices: D′(2, 4), E′(4, 1), F′(−2, −1). Since 0 < k < 1, this is a reduction — the image is half the size of the pre-image.

Visual — Dilation by k = 2



Rule: (x, y) → (2x, 2y)

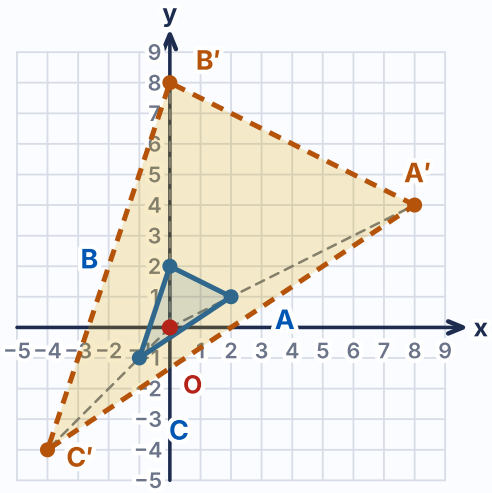
blue = pre-image · **gold dashed** = image

Scale factor k = 2 (enlargement): each image point is twice as far from O along the same ray.

Practice

P1 Triangle ABC has vertices A(2, 1), B(0, 2), C(−1, −1). The triangle is dilated by scale factor k = 4 (centered at origin). What are the new coordinates of A′?

- ● **(8, 4)**
- ○ (6, 5)
- ○ (2, 4)
- ○ (0.5, 0.25)

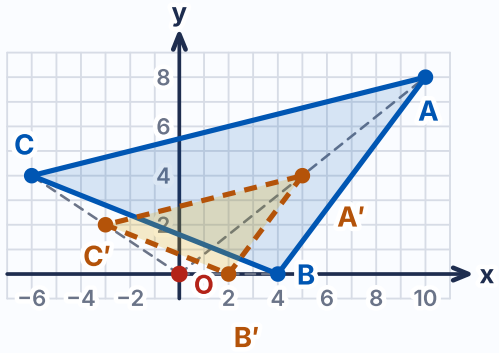


Rule: (x, y) → (4x, 4y)

blue = pre-image · **gold dashed** = image

P2 Triangle ABC has vertices A(10, 8), B(4, 0), C(−6, 4). The triangle is dilated by k = $\frac{1}{2}$ (centered at origin). Write the new coordinates of vertex A′:

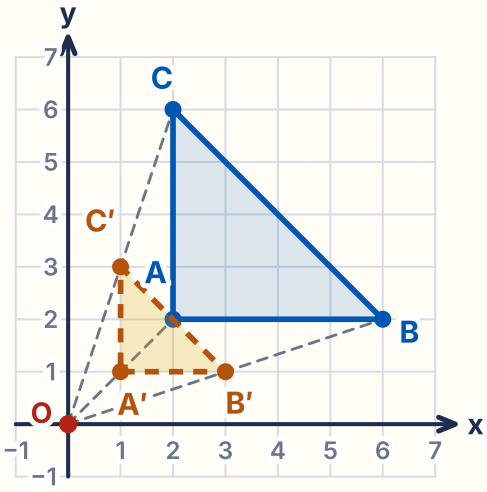
A′ = (5 , 4)



Rule: (x, y) → (½x, ½y)

blue = pre-image · **gold dashed** = image

P3 A figure is dilated and the image is *smaller* than the pre-image. The scale factor k must be...



The **image** is smaller. What must k be?

- ○ greater than 1
- ○ equal to 1
- ● **between 0 and 1**
- ○ negative

6 Congruence & Similarity

Every transformation we've studied tells us how two figures are related. Here's the big-picture connection:

📌 **Congruent figures (≅):** Same size AND same shape. One can be mapped to the other using only **rigid transformations** (translations, reflections, rotations — or any combination).

📌 **Similar figures (~):** Same shape but possibly different size. One can be mapped to the other using **rigid transformations AND a dilation**.

The Rule

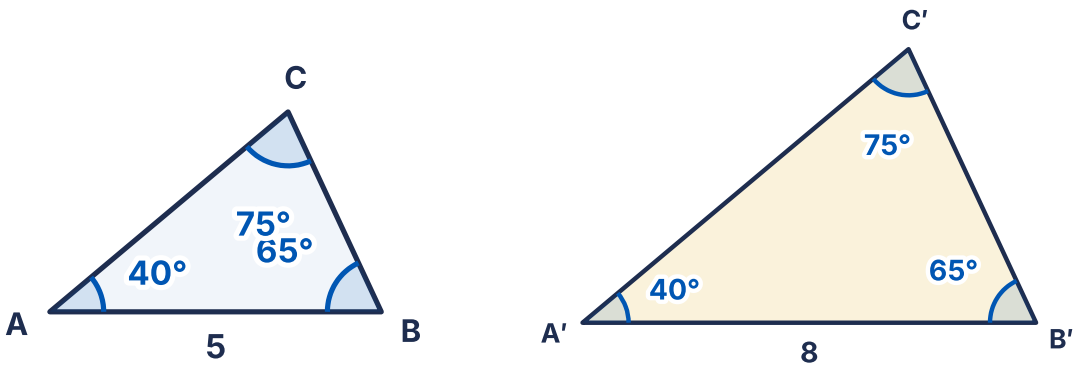
If you can get from Figure A to Figure B using **only translations, reflections, and rotations** → they are **CONGRUENT**.

If you need a **dilation** somewhere in the sequence → they are **SIMILAR** (but not congruent, unless k = 1).

What Gets Preserved?

Property	Translations	Reflections	Rotations	Dilations
Side lengths	✓ same	✓ same	✓ same	× scaled by k
Angle measures	✓ same	✓ same	✓ same	✓ same
Shape	✓ same	✓ same	✓ same	✓ same
Size	✓ same	✓ same	✓ same	× changes

💡 **Notice:** Dilations are the only transformation that changes *side lengths*. Angles stay the same in every transformation — that's why the shape never distorts.

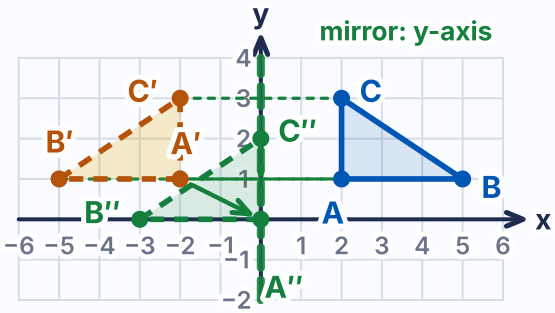


Dilation with k = 1.6: the 40°, 65° and 75° angles stay the same; side 5 becomes 5 × 1.6 = 8.

Worked Example

Example 6: Triangle ABC is reflected over the y-axis, then translated by (x + 2, y − 1). Are the pre-image and image congruent, similar, or neither?

A sample △ABC through the same two moves.



Step 1: (x, y) → (−x, y)
Step 2: (x, y) → (x + 2, y − 1)

blue → **gold** (step 1) → **green** (step 2)

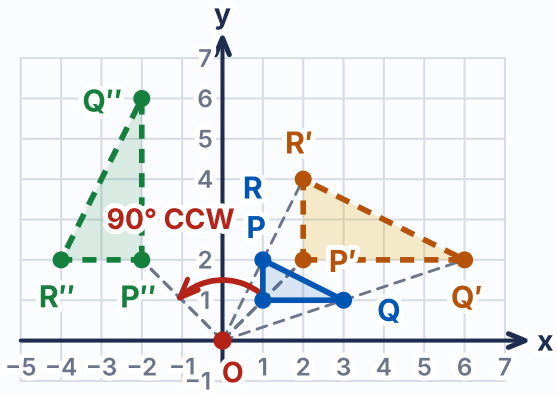
Step 1 — List the transformations used: Reflection over y-axis (rigid) + Translation (rigid).

Step 2 — Any dilations? No — both transformations are rigid (no resizing).

✓ **Conclusion: The figures are CONGRUENT. Side lengths and angles are all preserved.**

Example 7: Triangle PQR is dilated by k = 2, then rotated 90° CCW. Congruent, similar, or neither?

A sample △PQR through the same two moves.



Step 1: (x, y) → (2x, 2y)
Step 2: (x, y) → (−y, x)

blue → **gold** (step 1) → **green** (step 2)

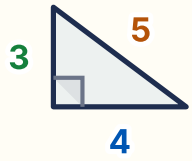
Step 1 — List the transformations: Dilation by k = 2 (non-rigid) + Rotation (rigid).

Step 2 — Was a dilation used (with k ≠ 1)? Yes — the dilation changes side lengths by a factor of 2.

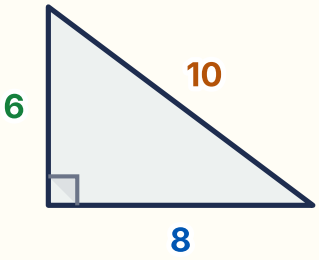
✓ **Conclusion: The figures are SIMILAR but NOT congruent. Same shape, different size.**

Practice

P1 Triangle A has side lengths 3, 4, 5. Triangle B has side lengths 6, 8, 10 and the same angles. Triangles A and B are...



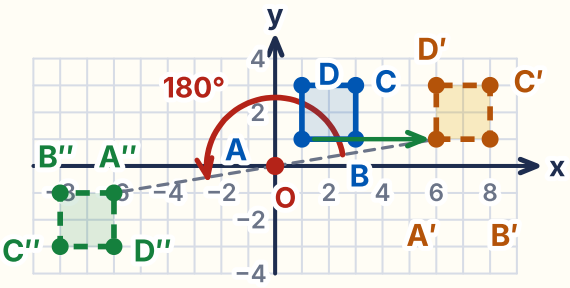
Triangle A



Triangle B

- ○ Congruent
- ○ Similar
- ○ Neither

P2 A square is translated 5 units right, then rotated 180° about the origin. The pre-image and image are...

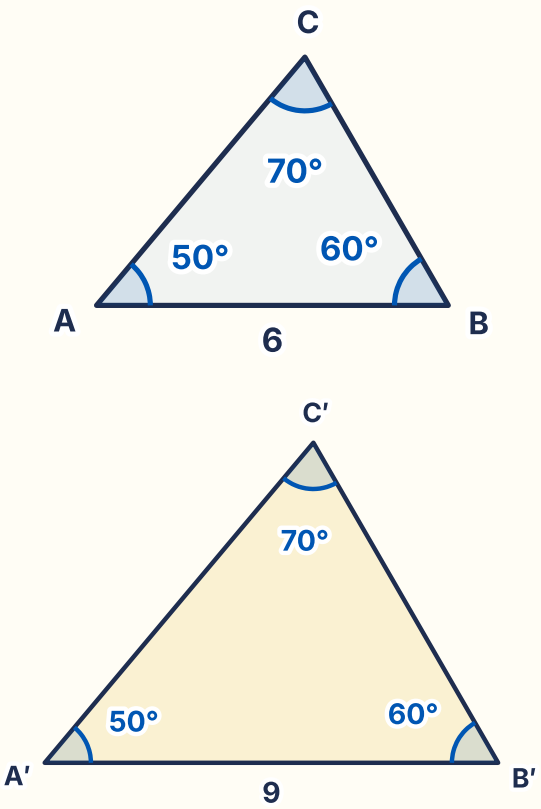


Step 1: slide 5 right
Step 2: turn 180°

blue → **gold** (step 1) → **green** (step 2)

- ○ Congruent
- ○ Similar but not congruent
- ○ Neither

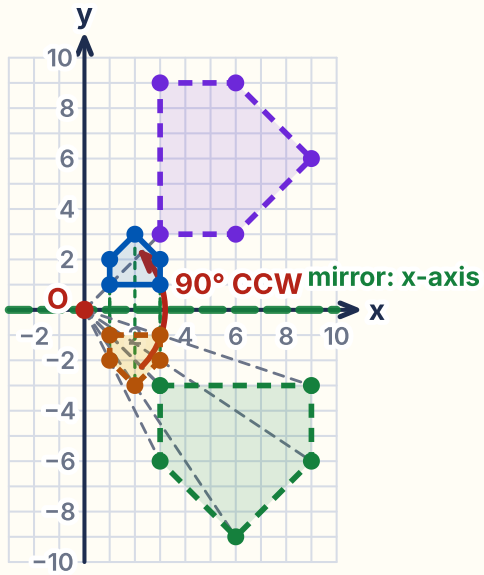
P3 **Which transformation preserves angle measure but NOT side length?**



After a dilation with k = 1.5, the angles are still 50°, 60° and 70°, but side 6 becomes 6 × 1.5 = 9.

- ○ Translation
- ○ Reflection
- ○ Rotation
- ○ Dilation

⭐ Challenge **A pentagon undergoes a sequence: reflection over the x-axis, dilation by k = 3, then rotation 90° CCW. The pre-image and image are...**



Step 1: flip over x-axis
Step 2: dilate k = 3
Step 3: turn 90° CCW

blue → **gold** (step 1) → **green** (step 2) → **purple** (step 3)

- ○ Congruent
- ○ Similar (not congruent)
- ○ Neither

Unit 8 Wrap-Up

You've now seen all four transformation types and how they connect to congruence and similarity. Quick recap:

- **Translation:** (x, y) → (x + h, y + k) — slide. Rigid.
- **Reflection over x-axis:** (x, y) → (x, −y) — y flips. Rigid.
- **Reflection over y-axis:** (x, y) → (−x, y) — x flips. Rigid.
- **Reflection over y = x:** (x, y) → (y, x) — swap. Rigid.
- **Rotation 90° CCW:** (x, y) → (−y, x). Rigid.

- **Rotation 180°:** (x, y) → (−x, −y). Rigid.
- **Rotation 90° CW (same as 270° CCW):** (x, y) → (y, −x). Rigid.
- **Dilation (k from origin):** (x, y) → (kx, ky). Non-rigid.

💡 **Next step:** Head to the Unit 8 page to try the practice problems and the Transformation Quest game.